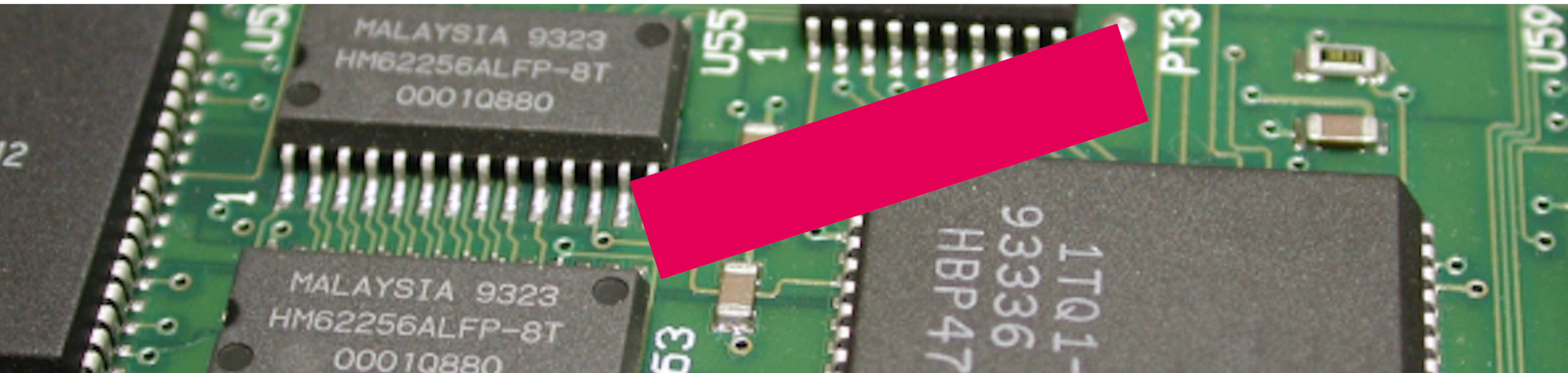


Embedded Systems Development - 7. Constructors, destructors, overriding and unit tests.



Electrical Engineering / Embedded Systems
School of Engineering and Automotive

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Schedule (exact info see #00 and roster at insite.han.nl)

Week (Thu)	Design principle	C++	UML	Lab (Fri)
Step 1 - 3 Sep	Single Responsibility	Scope, namespaces, string	Use cases / class diagram	4 Sep: Git
Step 2 - 10 Sep	Open-Closed Principle	Constructors, iterators, lambdas	Inheritance / generalization	11 Sep: Requirements
Step 3 - 17 Sep	Liskov Substitution Principle	Lists, inline functions, default params	Activity diagrams	18 Sep: sensor library 1: core, Bus, Arduino
Step 4 - 24 Sep	Interface Segregation Principle	Interfaces and abstract classes	Dependencies	25 Sep: sensor library 2: Raspberry Pi, Bus / Clock split
Step 5 - 1 Oct	Dependency Inversion Principle	Threads, callbacks	Sequence diagrams	2 Oct: sensor library 3: sampler thread, callback, MQTT
Step 6 - 8 Oct	Coupling and cohesion	Polymorphism	Composition, packages	9 Oct: sensor library 4: polymorphism, UML design
Step 7 - 15 Oct	Embedded SOLID	Lifecycle: constructors, destructors, const	-	16 Oct: sensor library 5: RAIL, host unit tests, refactor

Next period

Work on your project(s)

Peer reviewing of code

Feedback on diagrams

Feedback on Code

Please fill in the Excel I shared...

Please use https://github.com/AEAEmbedded/ESE_PROG

Exam preparation (deprecated), use this for final project...

On top of the topics from the previous slide (some items might be on both slides):

C++

- **constructor and destructor (RAII)**
- access (visibility: private / public / protected)
- inheritance: base class, derived class, virtual functions, **override**
- class / struct
- abstract / interface, Abstract Base Class, abstract member function = 0
- getter / setter
- **const-correct**
- composite and aggregate
- range based for-loop (foreach) for container classes
- std:: name space, std::vector

UML and concepts

- class diagram, composite and aggregate, inheritance
- stereotype / object / actor
- abstract class / interface
- package diagram
- sequence diagram
- polymorphism, polymorphic arrays and vectors
- use cases
- state diagrams, events and states (prior knowledge)

Note: topics from previous C/C++ courses are considered as existing prior knowledge.

Note: the five *SOLID* principles will not be explicitly asked during the exam but can be helpful during the exam and are considered standard practices for good software engineering.

C++ Language: Lifecycle / lifetime of objects

Typically a variable or object exists as long as it is “in scope”

```
#include <iostream>

class IamALife {
public:
    IamALife() {
        std::cout << "I was created ex nihilo!" << std::endl;
    }

    ~IamALife() {
        std::cout << "My lifecycle ended: IamALife is no more :(" << std::endl;
    }

    void shoutAloud() {
        std::cout << "Whooohoo, IamALife!" << std::endl;
    }
};

int main() {
    std::cout << "There we go..." << std::endl;
    IamALife toBeOrNotToBe;
    toBeOrNotToBe.shoutAloud();
    std::cout << "Main almost out of scope..." << std::endl;
}
```

C++ Language: Lifecycle / lifetime of objects

So default lifecycle:

- object is created in a certain scope...
- see the example if you don't or want to check if you understand scope

Note: lifetime is a *runtime* property of objects.

C++ Language: Lifecycle / lifetime of objects

So default lifecycle:

- object is created in a certain scope...
- see the example if you don't or want to check if you understand scope

But what if you create an object using new...

C++ Language: Lifecycle / lifetime of objects

Typically a variable or object exists as long as it is “in scope”: in case you use new this is not handled automatically anymore...

```
#include <iostream>
#include <vector>

class IamALife {
public:
    IamALife() {
        std::cout << "I was created ex nihilo!" << std::endl;
    }

    ~IamALife() {
        std::cout << "My lifecycle ended: IamALife is no more :(" << std::endl;
    }

    void shoutAloud() {
        std::cout << "Whoohoo, IamALife!" << std::endl;
    }
};

std::vector<IamALife *> _hamlets;

int main() {
    std::cout << "There we go..." << std::endl;

    IamALife *toBeOrNotToBe = new IamALife(); // requires delete!!!
    toBeOrNotToBe->shoutAloud();

    std::cout << std::endl;
    std::cout << "Main goes out of scope after this..." << std::endl;
    std::cout << "Whoops IamALife is becoming a zombie..." << std::endl;
    delete toBeOrNotToBe; // phew... just in time...
}
```

C++ Language: constructor and destructor (RAII)

Resource Acquisition Is Initialization (RAII):

- binds the life cycle of a resource to the lifetime of an object
- that must be acquired before use
- use cases: e.g. allocated heap memory, thread of execution, open socket, open file, locked mutex, disk space, database connection

This helps to prevent resource leaks and makes code exception-safe.

RAII, three times in your own code: LinuxI2cBus

```
class LinuxI2cBus final : public Bus {
public:
    LinuxI2cBus(const char* device, uint8_t address)
        : address_(address) {
        fd_ = ::open(device, O_RDWR);           // acquire
        if (fd_ >= 0 && ::ioctl(fd_, I2C_SLAVE, address_) < 0) {
            ::close(fd_); fd_ = -1;
        }
    }
    ~LinuxI2cBus() override {
        if (fd_ >= 0) ::close(fd_);             // release, always
    }
    bool isOpen() const { return fd_ >= 0; }
private:
    int fd_ = -1;
    uint8_t address_;
};
```

RAII, three times: Sampler and MqttPublisher

```
Sampler::~Sampler()                { stop(); }                // joins the thread

void Sampler::stop() {
    if (!running_) return;
    running_ = false;
    wakeUp_.notify_all();           // do not wait out the period
    if (thread_.joinable()) thread_.join();
}

MqttPublisher::MqttPublisher(host, port) {
    mosquitto_lib_init();
    client_ = mosquitto_new(...);  mosquitto_connect_async(...);  mosquitto_loop_start(client_);
}

MqttPublisher::~MqttPublisher() {
    mosquitto_disconnect(client_);  mosquitto_loop_stop(client_, false);
    mosquitto_destroy(client_);     mosquitto_lib_cleanup();
}

// A file descriptor, a thread, a network client: same shape every time.
```

Non-copyable: = delete, and why

```
LinuxI2cBus(const LinuxI2cBus&)          = delete;    // two objects, one fd: double close
LinuxI2cBus& operator=(const LinuxI2cBus&) = delete;

Sampler(const Sampler&)                  = delete;    // std::thread is not copyable anyway

MqttPublisher(const MqttPublisher&)      = delete;    // double mosquitto_destroy

Bme280(const Bme280&)                    = delete;    // the Bosch struct stores intf_ptr:
                                                    // a copy keeps pointing at the old bus

// Rule: a class that owns a resource is non-copyable.
// A class that is handed a reference (Bme280 gets Bus&) never deletes it.
```

Construction does not touch hardware

```
bme280::Bme280 sensor(bus, clock);    // cheap: stores two references, fills the struct

void setup() {
    Wire.begin();
    sensor.init();                     // now the chip is reset, ID checked, calibration read
}

// Why: on Arduino `sensor` is a global. Its constructor runs before setup(),
// before Wire.begin(), before Serial. Talking to the bus there fails silently.

// Same rule on the Pi: construct in main, init() once everything exists,
// and init() returns an Error you can act on. Constructors cannot.
```

Benefits of RAII

- Automatic Resource Management: RAII eliminates the need for manual resource management, reducing the risk of resource leaks (e.g., forgetting to close a file or release memory).
- Exception Safety: Even if exceptions are thrown, the destructor is still called, ensuring resources are released.
- Simplified Code: RAII leads to cleaner and more concise code by encapsulating resource management within objects.
- Reduced Errors: By automating resource management, RAII helps prevent common errors like memory leaks, double-frees, and dangling pointers.

Three Principles of RAI

- Encapsulate each resource into a class

The constructor of the class acquires the resource, and the destructor releases it.

- Instance ownership
- No manual release

Three Principles of RAI

- Encapsulate each resource into a class
- Instance ownership

Objects are usually instantiated on the stack or through smart pointers, ensuring deterministic destruction.

- No manual release

Three Principles of RAI

- Encapsulate each resource into a class
- Instance ownership
- No manual release

Avoid explicitly releasing resources.

Instead, rely on the destructor to do it automatically when the object goes out of scope.

Three Principles of RAI

In summary:

- Encapsulate each resource into a class
- Instance ownership
- No manual release

- Memory Management: Helps avoid memory leaks.
- Exception Safety: Ensures that resources are properly cleaned up if an exception is thrown.
- Simpler Code: Reduces the need for manual resource management.

C++ Language: Lifecycle / lifetime of objects

Strategies to avoid memory leaks:

- *garbage collection (e.g. Java) see: [https://en.wikipedia.org/wiki/Garbage_collection_\(computer_science\)](https://en.wikipedia.org/wiki/Garbage_collection_(computer_science))*
- *reference counting (C#, Swift, etc)*

In C++ one option is to use *smart pointers*...

C++ Language: constructor and destructor (RAII)

C++ does not perform reference-counting by default, fulfilling its philosophy of not adding functionality that might incur overheads where the user has not explicitly requested it. Objects that are shared but not owned can be accessed via a reference, raw pointer, or `iterator` (a conceptual generalisation of pointers).

However, by the same token, C++ provides native ways for users to opt-into such functionality: C++11 provides reference counted `smart pointers`, via the `std::shared_ptr` class, enabling automatic shared memory-management of dynamically allocated objects.

Programmers can use this in conjunction with `weak pointers` (via `std::weak_ptr`) to break cyclic dependencies. Objects that are dynamically allocated but not intended to be shared can have their lifetime automatically managed using a `std::unique_ptr`.

In addition, C++11's `move semantics` further reduce the extent to which reference counts need to be modified by removing the deep copy normally used when a function returns an object, as it allows for a simple copy of the pointer of said object.

C++ Language: Lifecycle / lifetime of objects

Using smart pointers: <https://www.youtube.com/watch?v=UOB7-B2MfwA>

```
#include <iostream>
#include <vector>
#include <memory>

class IamALife {
public:
    IamALife() {
        std::cout << "I was created ex nihilo!" << std::endl;
    }

    ~IamALife() {
        std::cout << "My lifecycle ended: IamALife is no more :(" << std::endl;
    }

    void shoutAloud() {
        std::cout << "Whoohoo, IamALife!" << std::endl;
    }
};

std::vector<IamALife *> _hamlets;

int main() {
    std::cout << "There we go..." << std::endl;
    std::cout << std::endl;

    { // local sub scope
        std::unique_ptr<IamALife> toBeOrNotToBe(new IamALife());
        toBeOrNotToBe->shoutAloud();
    }

    std::cout << std::endl;
    std::cout << "Main goes out of scope after this..." << std::endl;
}
```

C++ Language: Lifecycle / lifetime of objects

Using smart pointers: <https://www.youtube.com/watch?v=UOB7-B2MfwA>

```
// A. unique pointer:  
std::unique_ptr<IamAlife> toBeOrNotToBe = std::make_unique<IamAlife>();  
toBeOrNotToBe->shoutAloud();  
  
// B. is preferred over this:  
std::unique_ptr<IamAlife> toBeOrNotToBe(new IamAlife());  
toBeOrNotToBe->shoutAloud();
```

Reason to prefer option A over B is for exception purposes (A is safer / more robust against exceptions).

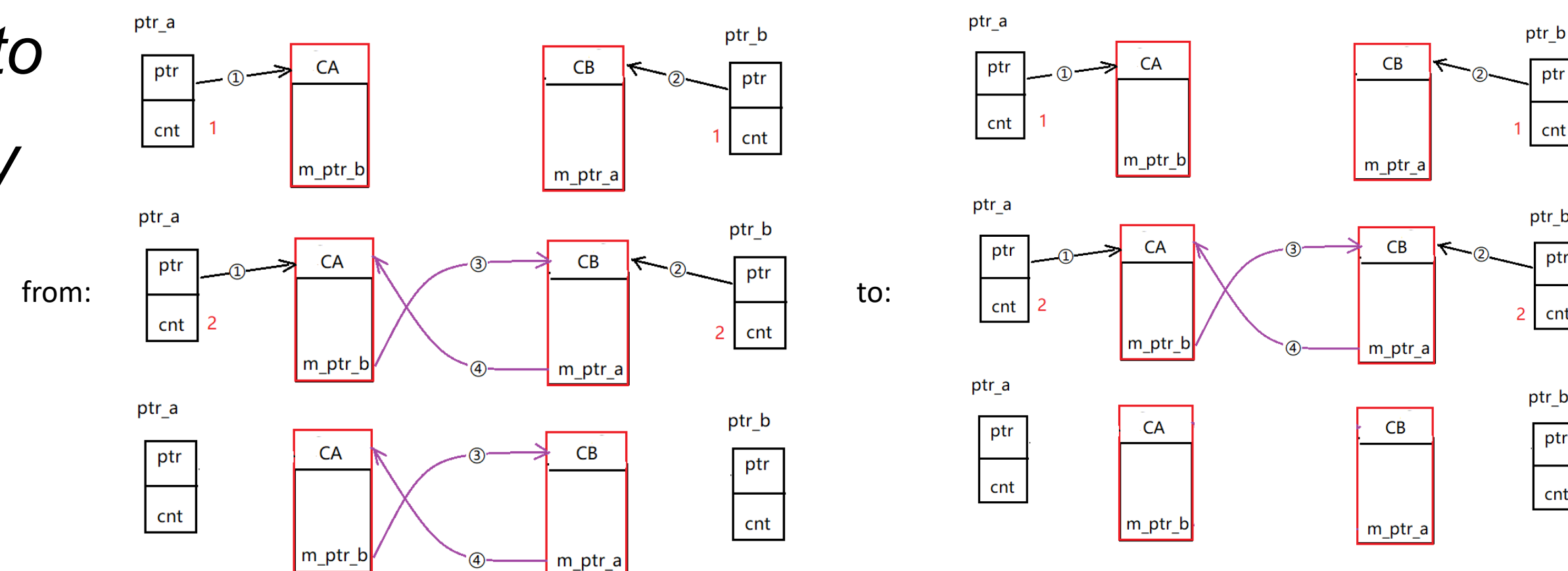
C++ Language: Lifecycle / lifetime of objects

Using smart pointers: <https://www.youtube.com/watch?v=UOB7-B2MfwA>

```
// Shared pointer:
std::unique_ptr<IamAlive> toBeOrNotToBe = std::make_shared<IamAlive>();
toBeOrNotToBe->shoutAloud();

// Weak pointer:
std::weak_ptr<IamAlive> weakPointer = toBeOrNotToBe;
weakPointer->shoutAloud();
```

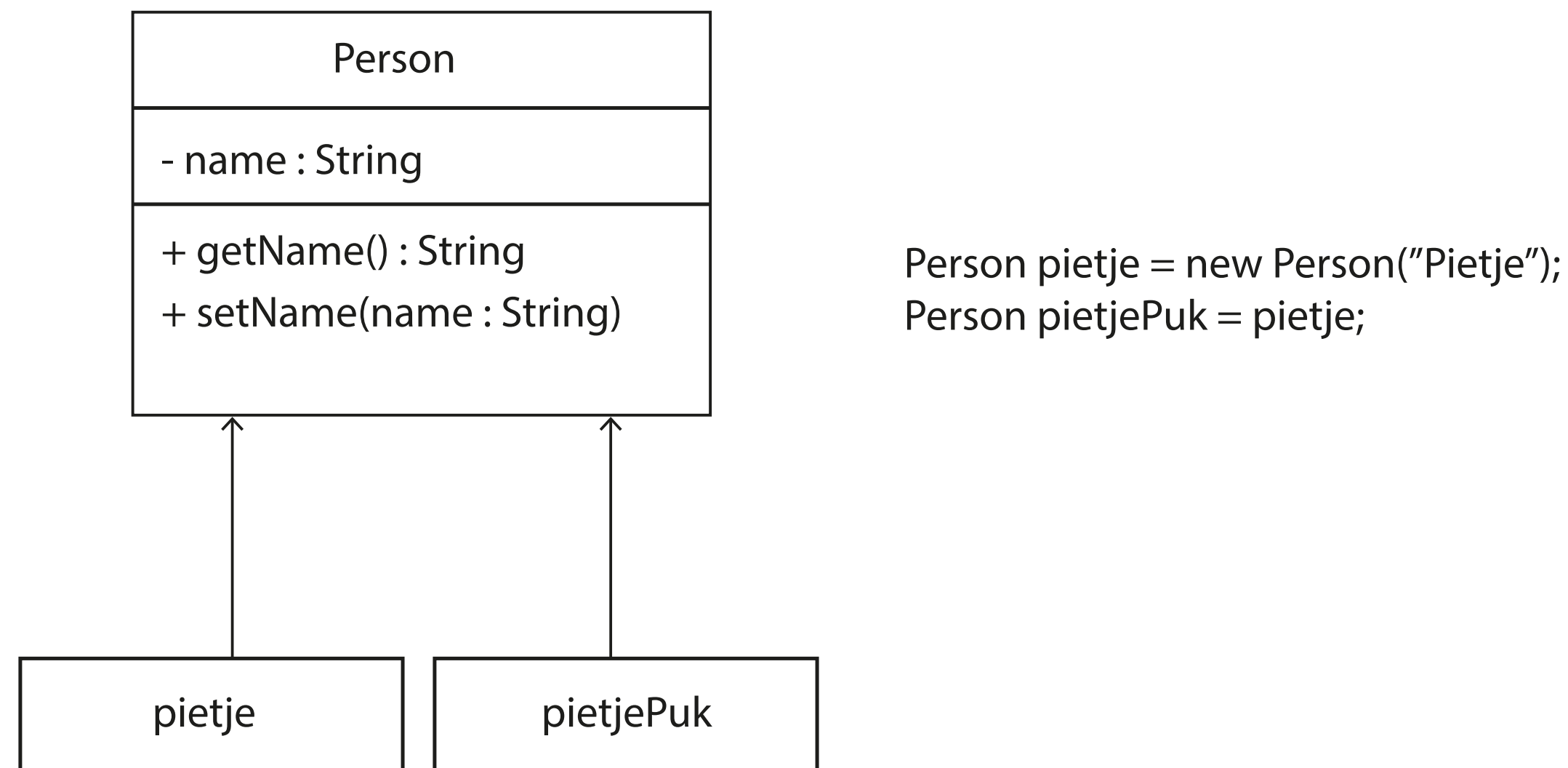
Weak helps reference counting to work properly and avoid memory leaks:



C++ Language: Lifecycle / lifetime of objects

Note: the following is of course very powerful (useful cases would include doubly linked lists).

This will not work when using unique pointers!



C++ Language: const-correctness (simple variables)

Const-correctness means *using the keyword const* to prevent const objects from getting mutated (in other words: making them *immutable*).

- the const keyword in front of a property
- `const int counter = 10;`
- remember: use a constructor to write only once to a const class attribute

C++ Language: const-correctness (arguments for methods)

Const-correctness means *using the keyword const* to prevent const objects from getting mutated.

Arguments (aka parameters) in methods:

- void f1(const std::string& s); // Pass by reference-to-const
- void f2(const std::string* sptr); // Pass by pointer-to-const
- void f3(std::string s); // Pass by value

C++ Language: const-correctness (const methods)

Const-correctness means *using the keyword const* to prevent const objects from getting mutated.

E.g. if you want to make a calculation using some function (= a method intended to give back a result):

```
int Loan::calcInterest() const
{
    return loan_value * interest_rate;
}
```

const-correctness on the sensor library

```
class Bme280 {
public:
    Error init(const Config& config = Config{});    // const&: not copied, not modified
    Error readForced(Measurement& out);             // out-parameter: not const
    uint32_t measurementTimeUs() const;            // asks nothing of the hardware
    bool initialised() const { return initialised_; }
};

using Callback = std::function<void(const Measurement&)>;    // observers may look, not touch

// Why can readForced() not be const?
// It changes the sensor's mode and the Bosch struct: observable state.
// Marking it const would be a lie to the caller (and to the compiler).
```

C++ Language: override

Override will allow to use the same method (including arguments and return) and override it's behavior in the same and/or subclasses.

```
struct A
{
    virtual void foo();
    void bar();
};

struct B : A
{
    void foo() override; // OK: B::foo overrides A::foo
};
```

C++ Language: override / shadowing

Override will allow to use the same method (including arguments and return) and override it's behavior in the same and/or subclasses.

```
struct Base
{
    virtual void foo();
    void bar();
    Base(); // constructor of Base
};

struct Derived : Base
{
    using Base::foo; // now foo method(s) from Base will become visible in Derived
    using Base::Base; // now we can use the constructor from Base in Derived
};
```

C++ Language: warning: shadowing/masking

You could get into trouble if you are not careful with overloading (inadvertently typecasting):

```
class Base {
public:
    int F(int i) { return i; };
};

class Derived : public Base {
public:
    //using Base::F;
    double F(double d) { return d; };
};

int main() {
    // Note if you forget using using Base::F, this will still work:
    Derived d = Derived();
    std::cout << std::to_string(d.F(10)) << std::endl;
}
```

First output with using Base::F; enabled:

Second output with code as shown above:

```
jakorten@mbp-van-ja-2221 Constructors % ./constructors
10
jakorten@mbp-van-ja-2221 Constructors % c++ constructors.cpp -o constructors
jakorten@mbp-van-ja-2221 Constructors % ./constructors
10.000000
jakorten@mbp-van-ja-2221 Constructors %
```

C++ Language: override

Making override explicit by adding *override* (C++11 and up) will allow the compiler to check the base class to see if there is a virtual function with this exact signature. And if there is not, the compiler will show an error.

```
struct A
{
    virtual void foo();
    void bar();
};

struct B : A
{
    void foo() const override; // Error: B::foo does not override A::foo
                               // (signature mismatch)
    void foo() override; // OK: B::foo overrides A::foo
    void bar() override; // Error: A::bar is not virtual
};
```

SOLID in an Embedded Context

SOLID principles in embedded systems may have (additional) challenges.

However it can significantly enhance the *design*, *scalability*, and *maintainability* of embedded software, leading to more **robust** and **reliable** embedded systems.

SOLID in an Embedded Context

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However it can significantly enhance the *design*, *scalability*, and *maintainability* of embedded software, leading to more **robust** and **reliable** embedded systems.

SOLID in an Embedded Context: Single Responsibility Principle (SRP)

Embedded Context:

Modularize your code:

- Write functions and modules with a single responsibility. For example, separate the code that handles hardware interfacing from the business logic.

Use clear and concise naming conventions:

- This makes it easy to understand what each module or function is responsible for, enhancing maintainability.

SOLID in an Embedded Context: Open/Closed Principle (OCP)

Embedded Context:

Leverage polymorphism for hardware abstraction:

- You can define a base class (or interface) for a hardware device (HAL), and then extend it for specific device implementations.
- This way, your code remains open for extension but closed for modification.

Use function pointers or callbacks:

- This allows you to change behavior without modifying existing code.

(e.g. use patterns including Dependency Injection)

SOLID in an Embedded Context: Liskov Substitution Principle (LSP)

Embedded Context:

Ensure compatibility of derived classes:

- If using object-oriented programming, make sure that derived classes can be used in place of their base classes without causing unexpected behavior.

Carefully manage hardware dependencies:

- Ensure that derived classes do not introduce hardware dependencies that could break the substitutability principle.

SOLID in an Embedded Context: Interface Segregation Principle (ISP)

Embedded Context:

Provide device-specific interfaces:

- For hardware drivers and peripherals, offer interfaces tailored to the specific functionalities each component uses, rather than a monolithic interface.
- Minimize dependencies: Ensure that each module or component only includes the minimal interface it needs, reducing the footprint and complexity.

SOLID in an Embedded Context: Dependency Inversion Principle (DIP)

Embedded Context:

Use abstract interfaces for hardware components:

- This allows high-level modules to depend on abstractions rather than concrete implementations, facilitating testing and system modifications.

Employ dependency injection:

- Pass dependencies (e.g., hardware interfaces) as parameters to functions or objects, allowing you to easily substitute different implementations for testing or system configuration.

(e.g. use patterns including Dependency Injection)

SOLID in an Embedded Context: Considerations

- Memory Constraints
- Performance Constraints
- Testing and Simulation
- Documentation and Code Comments

SOLID in an Embedded Context: Considerations

- **Memory Constraints**

Be mindful of the memory overhead introduced by abstractions and object-oriented features.
Optimize data structures and minimize dynamic memory allocation.

- Performance Constraints

- Testing and Simulation

- Documentation and Code Comments

SOLID in an Embedded Context: Considerations

- Memory Constraints

- **Performance Constraints**

Evaluate the performance impact of abstractions and interface layers, ensuring that the system meets its real-time requirements.

- Testing and Simulation

- Documentation and Code Comments

SOLID in an Embedded Context: Considerations

- Memory Constraints
- Performance Constraints
- **Testing and Simulation**

Use the abstraction layers to facilitate unit testing and hardware-in-the-loop simulation by substituting real hardware interfaces with mock or simulated implementations.

- Documentation and Code Comments

SOLID in an Embedded Context: Considerations

- Memory Constraints
- Performance Constraints
- Testing and Simulation
- **Documentation and Code Comments**

Maintain thorough documentation and inline comments to help developers understand the design principles and interfaces, contributing to easier maintenance and enhancements.

(May I suggest you to use Doxygen. In most cases you could also let a LLM do the documentation but it is not always possible (or allowed))

Embedded SOLID: what does an abstraction cost?

Every interface costs something on a SAMD21 or an AVR:

- a vtable per class (flash) and a vpointer per object (RAM)
- an indirect call per read()/write(): a few cycles per byte
- no inlining across the interface boundary

Measure it, do not guess: build the Arduino example twice

arm-none-eabi-size / avr-size, with Bus as interface vs Arduinol2cBus hard-wired
report flash and RAM, with and without

Verdict for this library: acceptable. You bought host tests, a second platform and a fake sensor for a few hundred bytes.

When it is not acceptable: an ISR path, a per-sample loop at kHz rates. Prog 6: hardware proxy, humble object, templates instead of virtuals.

Introduction to testing.

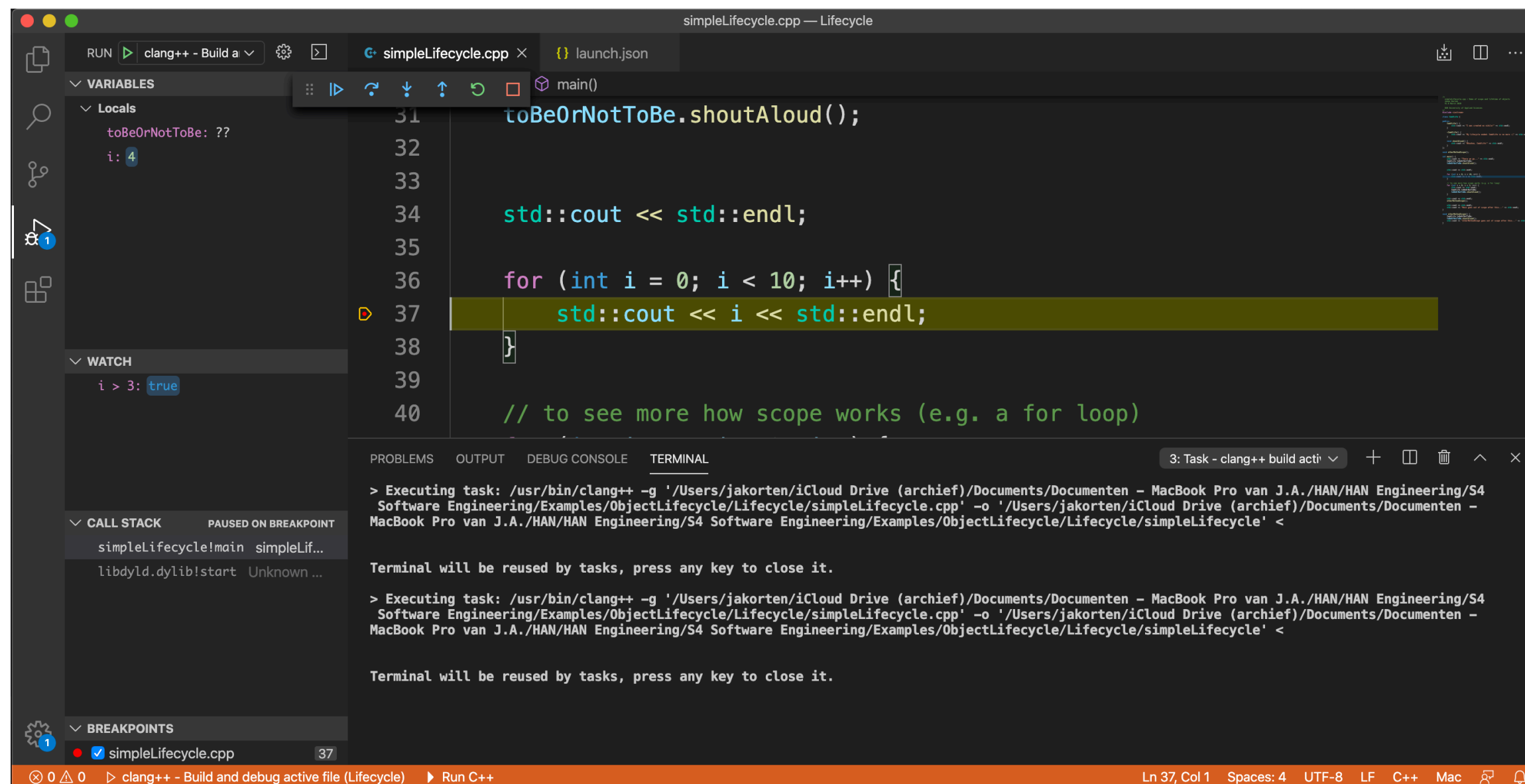
First of all: What is the difference between debugging and testing?

Introduction to testing in VS Code

Debugger in VS Code:

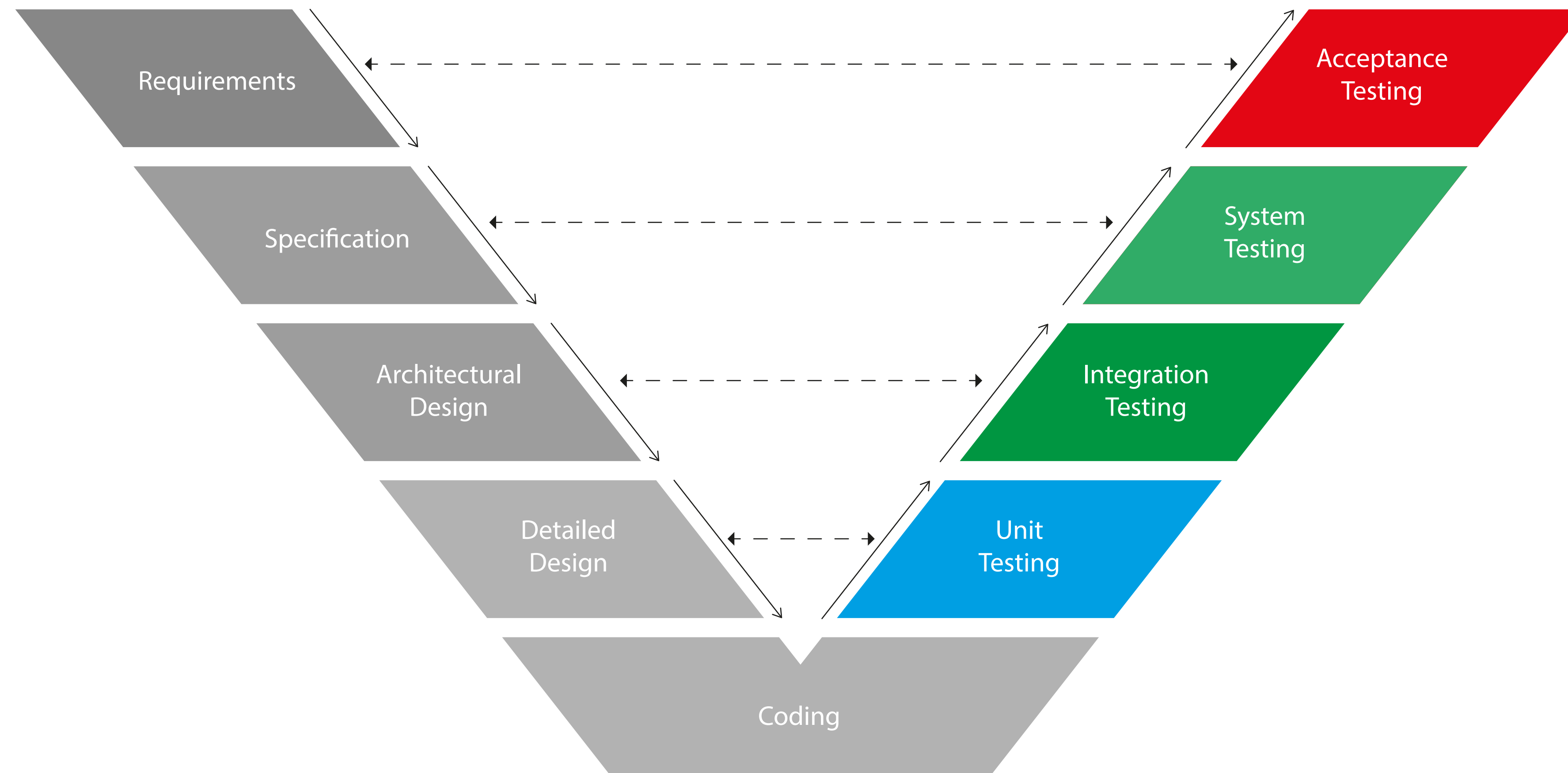
- <https://code.visualstudio.com/docs/cpp/launch-json-reference>
- <https://code.visualstudio.com/docs/editor/debugging>

Breakpoints, locals and watches:



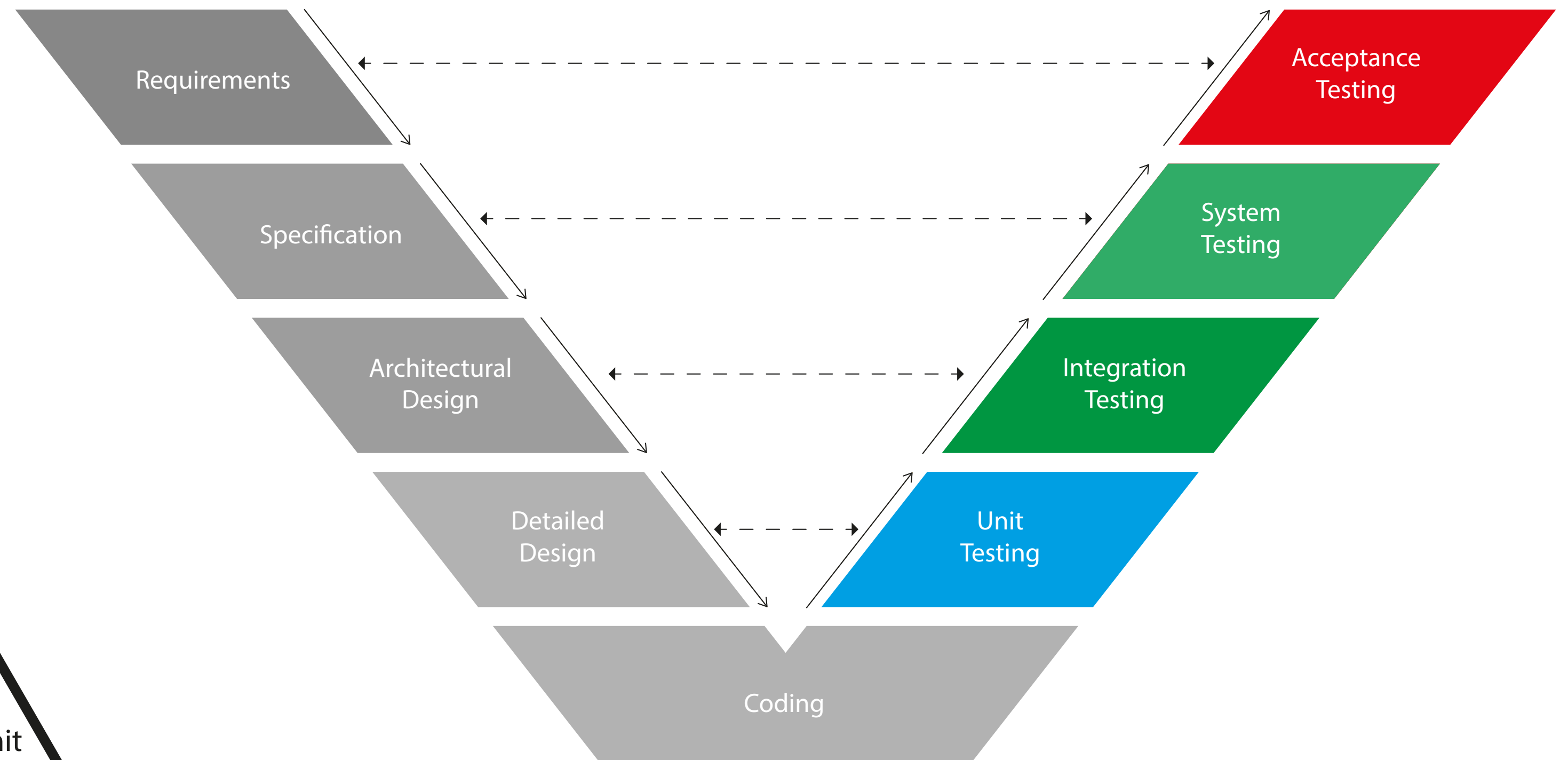
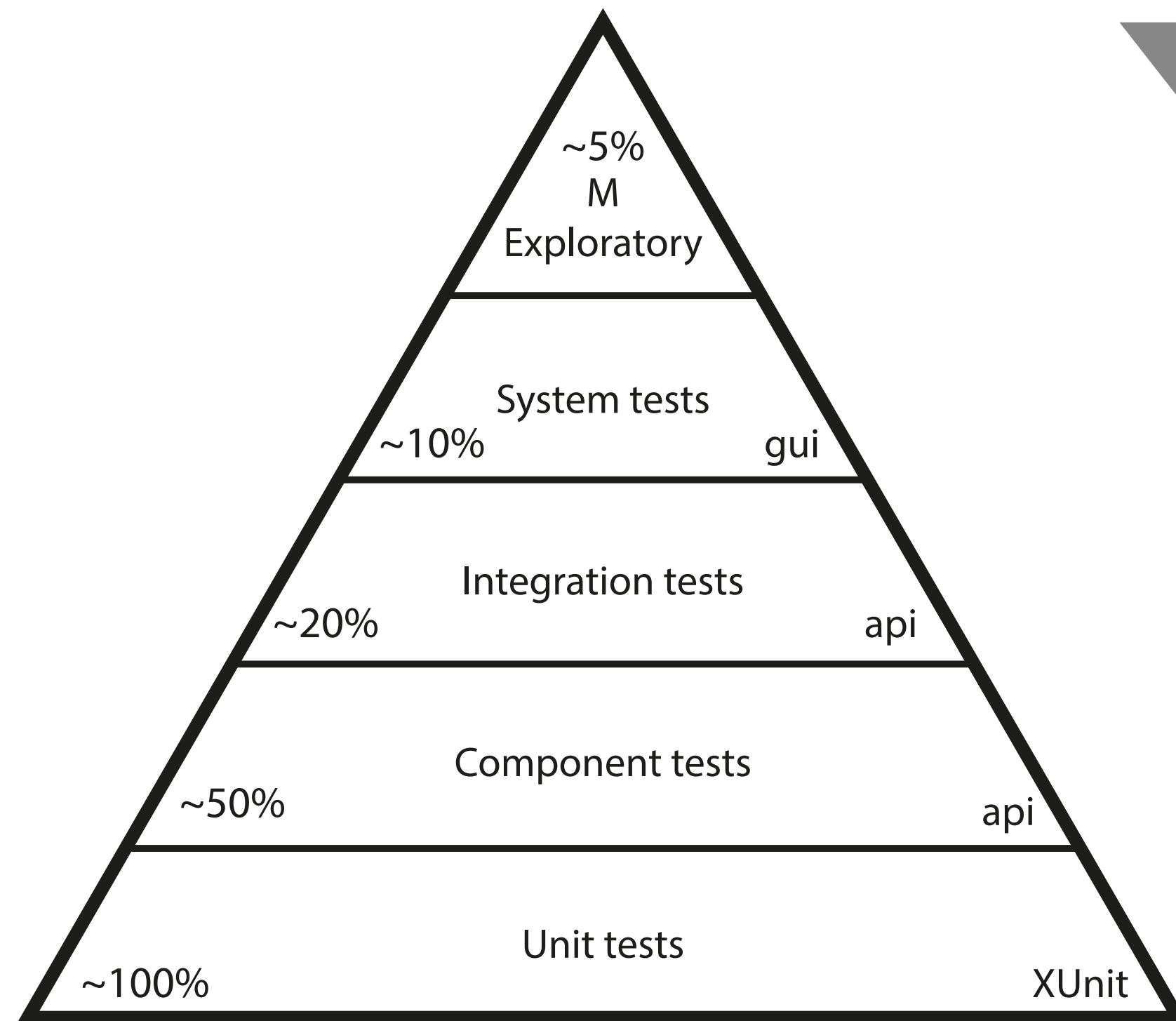
Note: (at least on macOS) it seems to require clang++ and llvm

Testing in an Embedded Systems Software Engineering context



Testing

Martin (2011, p.115)
provides us with
the “Test Automation Pyramid”:



Testing: Basic Unit testing in C++

For this period we will only introduce Unit testing.

Some basic rules for Unit tests:

- Testing uses exactly one assert per test.
- Avoid if-statements.
- Unit tests only “new()” the unit under test.
- Unit tests do not contain hard-coded values unless they have a specific meaning.
- Unit tests are stateless.
- Unit test should be fallible (read: are only useful if they can fail in the first place).

Testing: Basic Unit testing in C++

<https://code.visualstudio.com/api/working-with-extensions/testing-extension>

<https://marketplace.visualstudio.com/items?itemName=drleq.vscode-cpputf-test-adapter>

Note: For some reason I need to add `#include <cmath>` to CppUnitTestFramework.hpp

Testing: Basic Unit testing in C++

```
#define GENERATE_UNIT_TEST_MAIN
#include "CppUnitTestFixture.hpp"

#include <iostream>

class TestMe {

public:
    const int age = 10;

    TestMe() {
        std::cout << "I was created ex nihilo!" << std::endl;
    }

    ~TestMe() {
        std::cout << "My lifecycle ended: TestMe is no more :(" << std::endl;
    }
};

TestMe iLoveTesting;

TEST_CASE(TestMe, Test1) {

    CHECK_EQUAL(age, 9);

}
```

Unit tests on the host: MockBus and FakeClock

```
TEST_CASE("init fails on wrong chip id") {
    MockBus bus;  FakeClock clock;
    bus.reg(0xD0) = 0x00;                      // not 0x60 (BME280) / 0x58 (BMP280)
    Bme280 sensor(bus, clock);
    CHECK(sensor.init() == Error::WrongChipId);
    CHECK_FALSE(sensor.isInitialised());
}

TEST_CASE("readForced waits for the measurement") {
    MockBus bus;  FakeClock clock;              // FakeClock only counts
    Bme280 sensor(bus, clock);
    sensor.init();
    Measurement m;  sensor.readForced(m);
    CHECK(clock.delayedUs() >= sensor.measurementTimeUs());
}

// No hardware, no Pi, runs in ctest on your laptop. This is what the Bus / Clock split bought you.
```

The test that matters: the datasheet's worked example

```
TEST_CASE("compensation matches the datasheet example") {
    MockBus bus;  FakeClock clock;
    loadCalibration(bus, kDatasheetCalib);      // dig_T1..dig_H6 into 0x88.. and 0xE1..
    loadRawSample(bus, kDatasheetRaw);          // 0xF7..0xFE
    Bme280 sensor(bus, clock);
    REQUIRE(sensor.init() == Error::None);

    Measurement m;
    REQUIRE(sensor.read(m) == Error::None);
    CHECK(m.temperatureC == doctest::Approx(25.08).epsilon(0.01));
    CHECK(m.pressurePa    == doctest::Approx(100653).epsilon(0.001));
}

// Catches: endianness, sign of dig_T2/dig_P2, the 32-bit vs double macro, units (0.01 C, Pa, 1/1024 %RH).
// One test, most of the bugs students actually ship.
```

Assignments 7 and 8: tests, lifecycle, refactor (deadline Thu 22 Oct, tag v1.0)

7. Unit tests on the host, all run by ctest: wrong chip ID, bus failure, read before init, init with calibration, readForced waits via Clock, datasheet compensation. MqttPublisher is not unit-tested: say why, test Sampler with FakeSensor + RecordingPublisher instead.

8. Review at tag v0.6-design against DRY, KISS, SOLID, coupling / cohesion and the lifecycle checklist. Refactor and report per finding: what, where, which principle, what changed, which commit.

Mandatory refactor: add a second sensor (SHT45, or FakeSensor). Does it depend on bme280::? Then Bus, Clock and EnvironmentSensor move to their own package (hal). Package diagram before and after; Sampler and publishers must not change.

Plus one paragraph with real numbers: what the interfaces cost in flash and RAM.

Full hand-out, starter code and tags: github.com/AEAEmbedded/ESE_PROG -> Prog5/Assignments

One more thing:

Remember to use doxygen for your documentation.

“

Any questions?